

CS5340

Uncertainty Modeling in AI

Lecture 7: Mixture Models and the EM Algorithm

Assoc. Prof. Lee Gim Hee

AY 2026/27

Semester 1

Course Schedule

Week	Date	Topic	Remarks
1	10 Aug	National Day Holiday	No Lecture
2	17 Aug	Introduction to probabilistic reasoning	Assignment 0: Python Numpy Tutorial (Ungraded)
	22 Aug	Bayesian networks (Directed graphical models)	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
3	24 Aug	Markov random Fields (Undirected graphical models)	
4	31 Aug	Variable elimination and belief propagation	Assignment 1: Belief propagation and maximal probability (15%) Assignment 1: Due (13 Sep, 2359 HRS)
	05 Sept	Factor graph and the junction tree algorithm	Make-up Lecture: Sat, 9.30am-12.30pm (LT15, Recording will be provided)
5	07 Sep	I will be traveling	No Lecture
6	14 Sep	Parameter learning with complete data	Assignment 2: Junction tree and parameter learning (15%) Assignment 2: Due (27 Sep, 2359 HRS)
-	21 Sep	Recess week	No Lecture
7	28 Sep	Mixture models and the EM algorithm	
8	05 Oct	Hidden Markov Models (HMM)	Assignment 3: Hidden Markov model (15%) Assignment 3: Due (18 Oct, 2359 HRS)
9	12 Oct	Monte Carlo inference (Sampling)	
10	19 Oct	Variational inference	Assignment 4: MCMC Sampling (15%) Assignment 4: Due (01 Nov, 2359 HRS)
11	26 Oct	Variational Autoencoder and Diffusion Models	
12	02 Nov	Graph-cut and alpha expansion	
13	09 Nov	Diwali/Deepavali	No Lecture

Final Exam: 21 Nov 2026

Acknowledgements

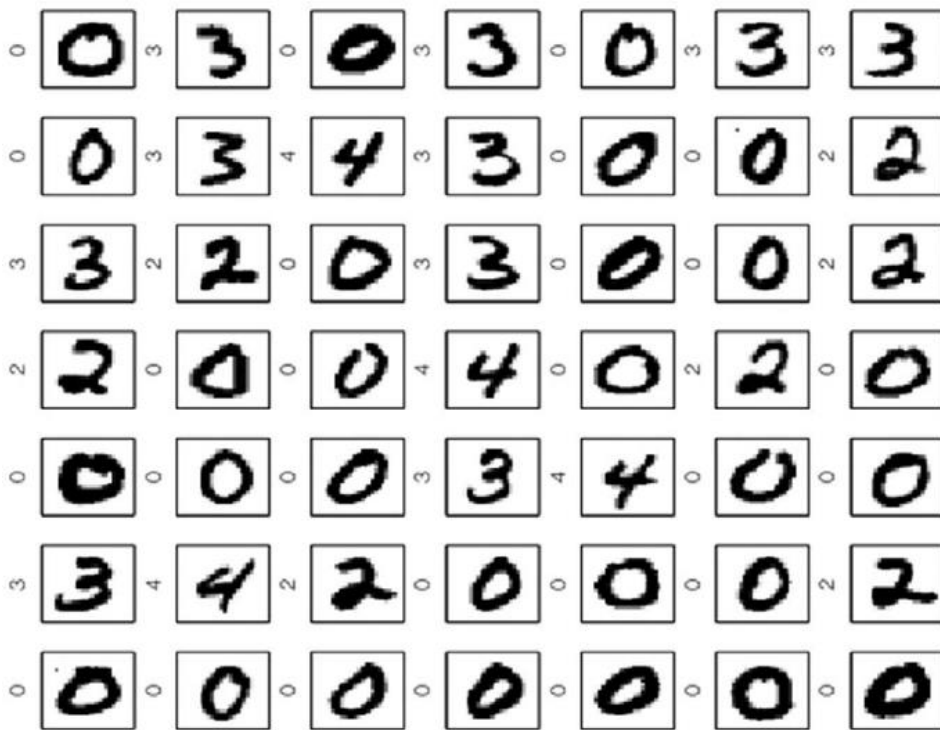
- A lot of slides and content of this lecture are adopted from:
 1. “Pattern Recognition and Machine Learning”, Christopher Bishop, Chapter 8.
 2. “Machine Learning – A Probabilistic Perspective”, Kevin Murphy, Chapter 11.
 3. “An Introduction to Probabilistic Graphical Models”, Michael I. Jordan, Chapters 10 and 11.
<http://people.eecs.berkeley.edu/~jordan/prelims/chapter10.pdf>
<http://people.eecs.berkeley.edu/~jordan/prelims/chapter11.pdf>
 4. “Probabilistic Graphical Models”, Daphne Koller and Nir Friedman, chapter 19.2.2.
 5. “Computer Vision: Models, Learning and Inference”, Simon Prince, Chapters 7.1-7.4 and 7.8.

Learning Outcomes

- Students should be able to:
 1. Use the non-probabilistic **k-mean algorithm** to solve the clustering problem.
 2. Describe the **Gaussian-mixture model**.
 3. Apply the **Expectation-Maximization algorithm** for estimation of both the unknown parameters and latent variables.

Motivation

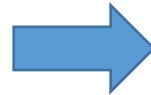
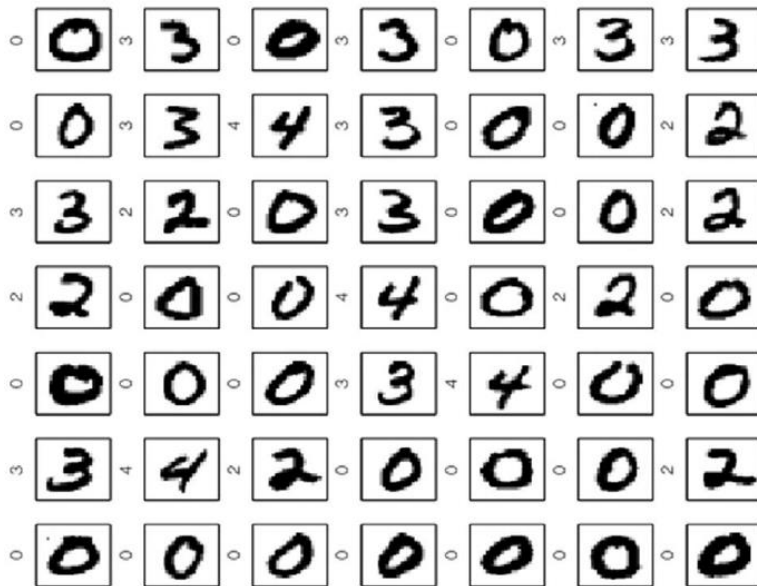
- Suppose that we are given **many images of handwritten digits**, and we can extract a 2D representation of each image $x_n \in \mathbb{R}^2$.



$x_1, x_2, x_3, \dots, x_N$

Motivation

- We can observe **clusters** from the 2D plot of $x_1, \dots, x_N$, where $x_n \in \mathbb{R}^2$.



$x_1, x_2, x_3, \dots, x_N$

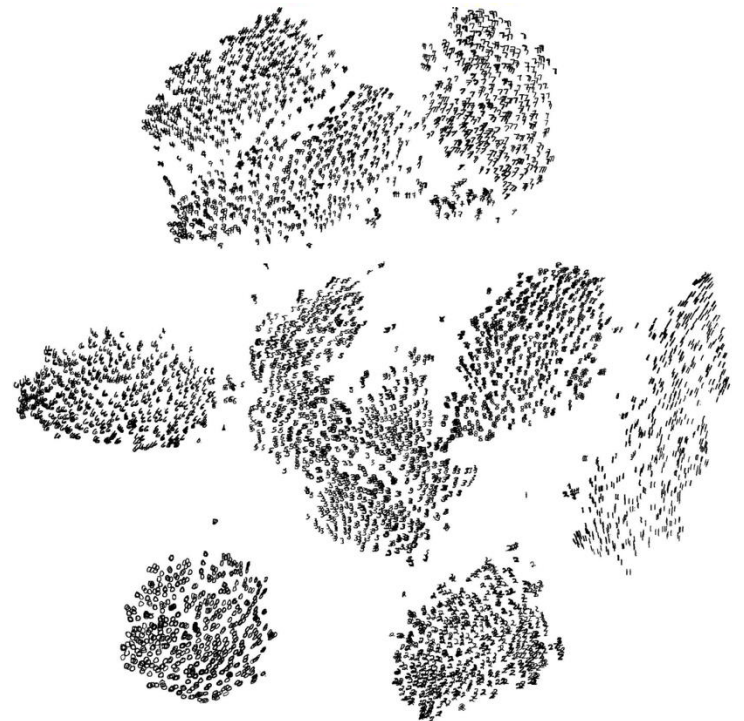
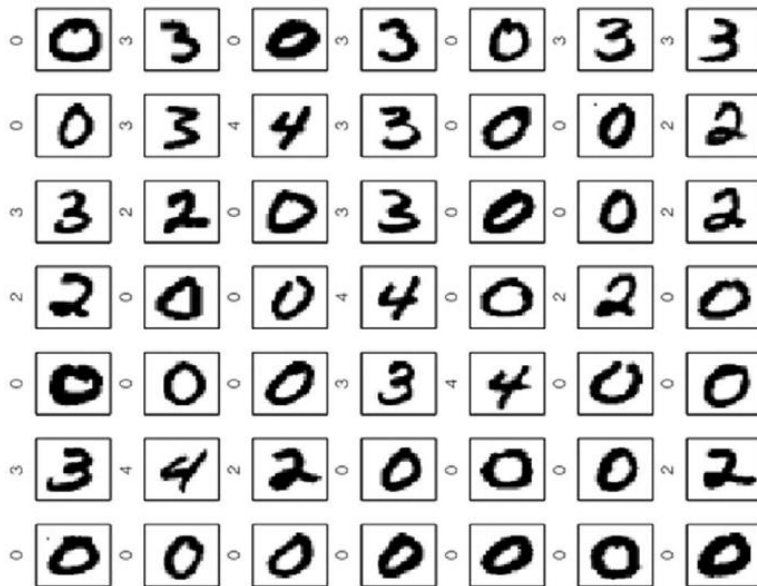


Image source: <https://stab-iitb.org/electronics-club/blog/2016/05/deep-learning-based-image-classification/>

Motivation

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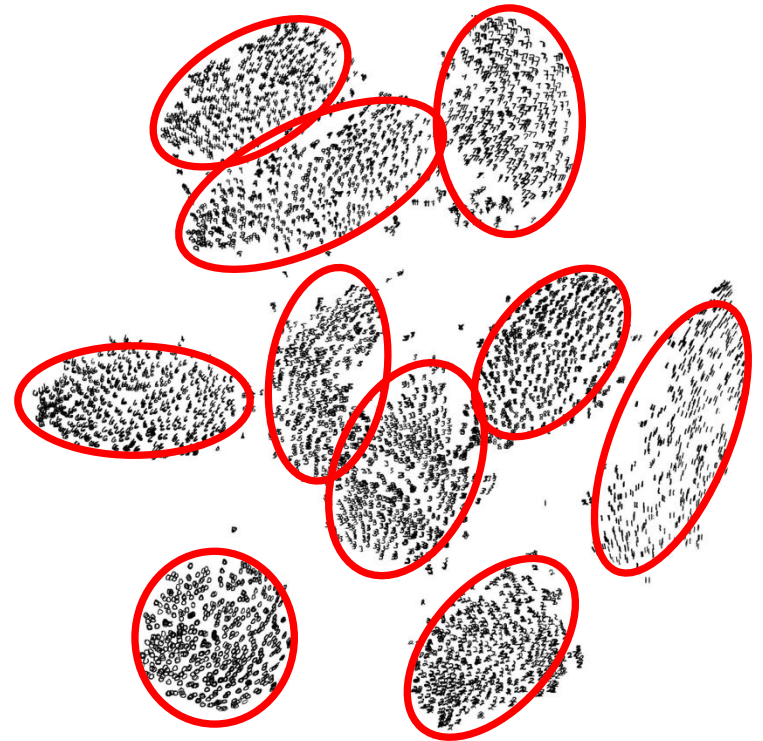


Image source: <https://stab-iitb.org/electronics-club/blog/2016/05/deep-learning-based-image-classification/>

Motivation

- It turns out that the data points x_n with the same digit tend to be **associated with the same cluster**!
- This suggests that clusters can be **used to represent complex distributions**.

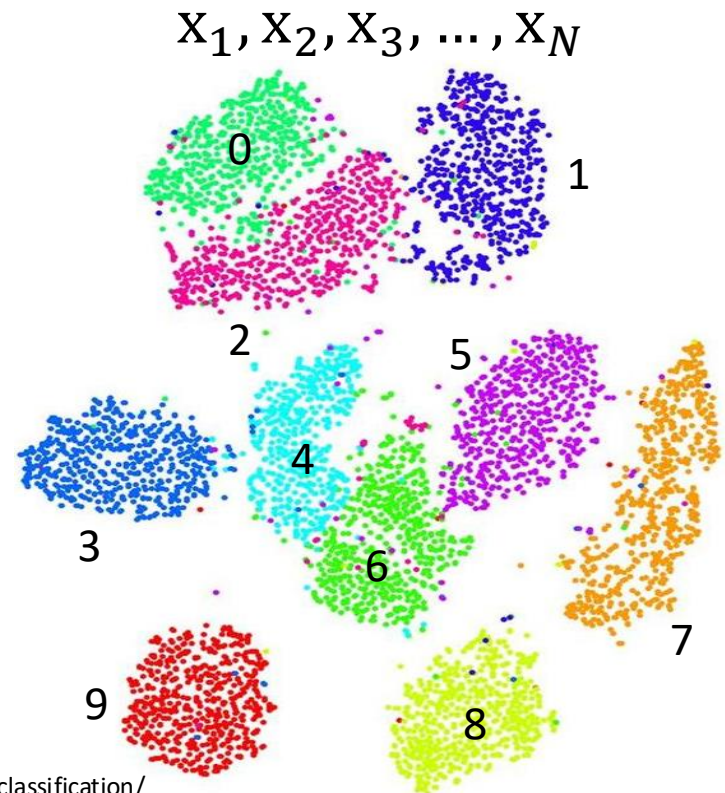
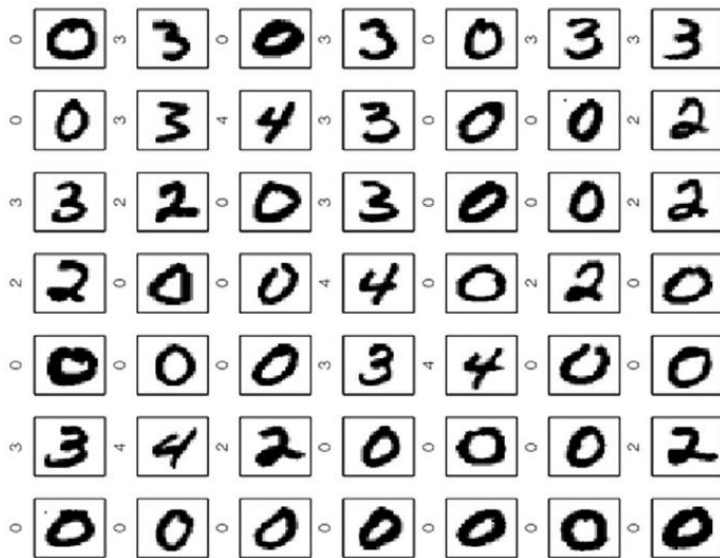


Image source: <https://stab-iitb.org/electronics-club/blog/2016/05/deep-learning-based-image-classification/>

Motivation

Chicken-and-Egg Problem:

1. How to find **unknown parameters** of the clusters?
2. Which cluster does each data belong to, i.e. **data association**?

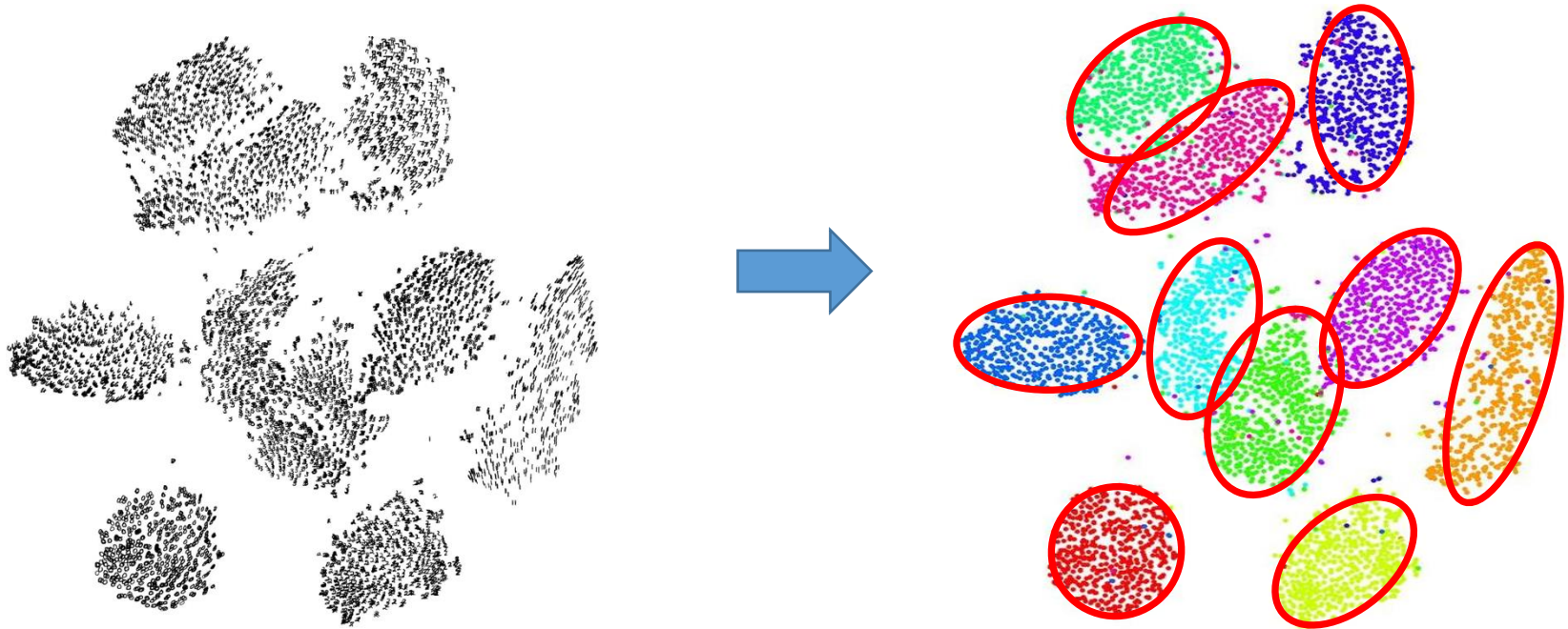


Image source: <https://stab-iitb.org/electronics-club/blog/2016/05/deep-learning-based-image-classification/>

Non-Probabilistic Approach: K-Means

Given:

1. **Data set** $\{x_1, \dots, x_N\}$ of observations
2. **Number of clusters** K

Find:

1. The K **cluster centers** $\{\mu_1, \dots, \mu_K\}$, i.e., unknown parameters (assume each cluster is a circle)
2. **Assignment of each point** x_n to a cluster center k , i.e., data association

Non-Probabilistic Approach: K-Means

1-of- K coding:

- For each data point x_n , we introduce a corresponding set of **binary indicator variables**:

$$r_{nk} \in \{0,1\} \quad \forall k = 1, \dots, K, \quad \text{s.t.} \quad \underbrace{\sum_k r_{nk}}_{\text{1-of-}K \text{ constraint}} = 1.$$

- This binary indicator variable describes which of the K clusters the data point x_n is assigned to.
- 1-of- K constraint ensures that each data point x_n gets **assigned to only ONE cluster k** .

Non-Probabilistic Approach: K-Means

- Formally, the **goal** of k-means is to find values for $\{r_{nk}\}$ and $\{\mu_k\}$ to **minimize the “distortion measure”**:

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \mu_k\|^2, \quad \text{s.t.} \quad \sum_k r_{nk} = 1.$$

- Represents the **sum-of-squares** of the distances of each point $\mathbf{x}_n$ to its assigned vector μ_k .

Non-Probabilistic Approach: K-Means

K-Means (a.k.a. Lloyd) Algorithm:

1. **Initialization**: Randomly choose some initial values for $\{\mu_k\}$.
2. **Assignment step**: Minimize J w.r.t. $\{r_{nk}\}$, while keeping $\{\mu_k\}$ fixed.
3. **Update step**: Minimize J w.r.t. $\{\mu_k\}$, while keeping $\{r_{nk}\}$ fixed.



Iterate until
convergence

Non-Probabilistic Approach: K-Means

Assignment Step:

- We can **optimize each r_n separately** since they are independent:

$$\operatorname{argmin}_{r_n} \sum_k r_{nk} \|\mathbf{x}_n - \mu_k\|^2, \quad \text{s.t.} \quad \sum_k r_{nk} = 1.$$

- By inspection, the minimum occurs when we assign $\mathbf{x}_n$ to the **current closest cluster center**:

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_n - \mu_j\|^2 \\ 0 & \text{otherwise.} \end{cases}$$

Non-Probabilistic Approach: K-Means

Update Step:

- We can **optimize each μ_k separately** since they are independent:

$$\operatorname{argmin}_{\mu_k} \underbrace{\sum_n r_{nk} \|\mathbf{x}_n - \mu_k\|^2}_L$$

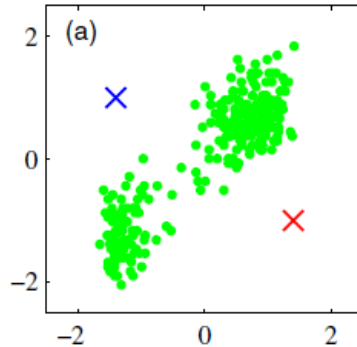
- Differentiation of L w.r.t. μ_k , and equate to zero gives:

$$2 \sum_{n=1}^N r_{nk} (\mathbf{x}_n - \mu_k) = 0 \quad \Rightarrow \quad \boxed{\mu_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}}}$$

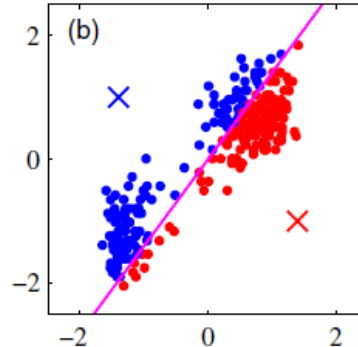
Mean of all points assigned to cluster k , hence “k-means”!

Non-Probabilistic Approach: K-Means

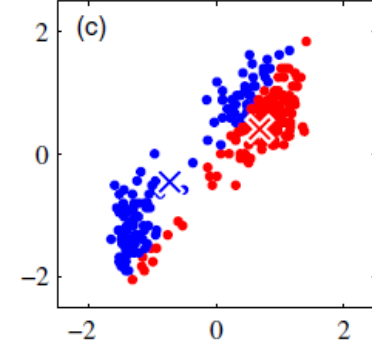
Initialization ($K = 2$):
Choose μ_k



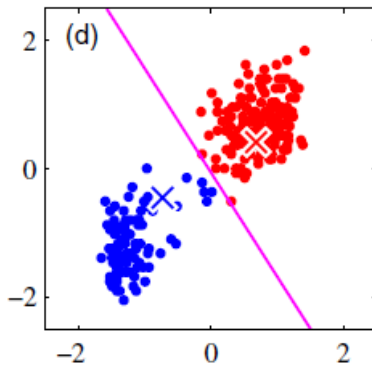
Iteration (1):
Assignment of r_{nk}



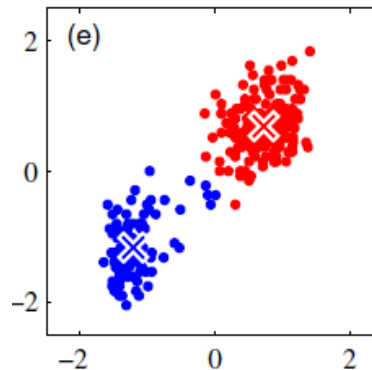
Iteration (1):
Update of μ_k



Iteration (2):
Assignment of r_{nk}



Iteration (2):
Update of μ_k



Iteration (3):
Assignment of r_{nk}

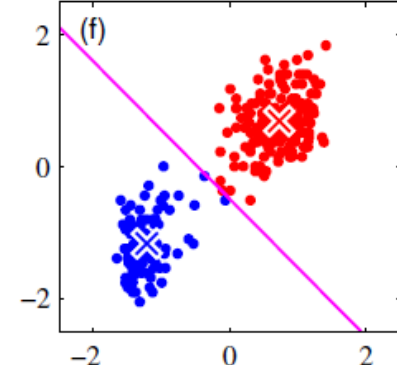
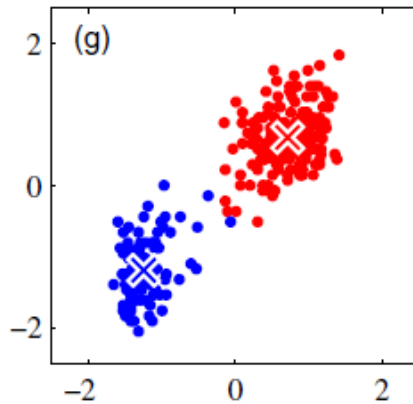


Image source: "Pattern recognition and machine learning", Christopher Bishop

Non-Probabilistic Approach: K-Means

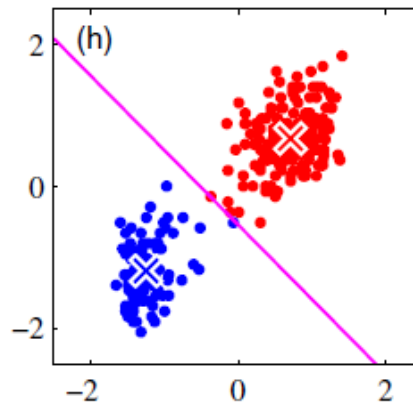
Iteration (3):

Update of μ_k



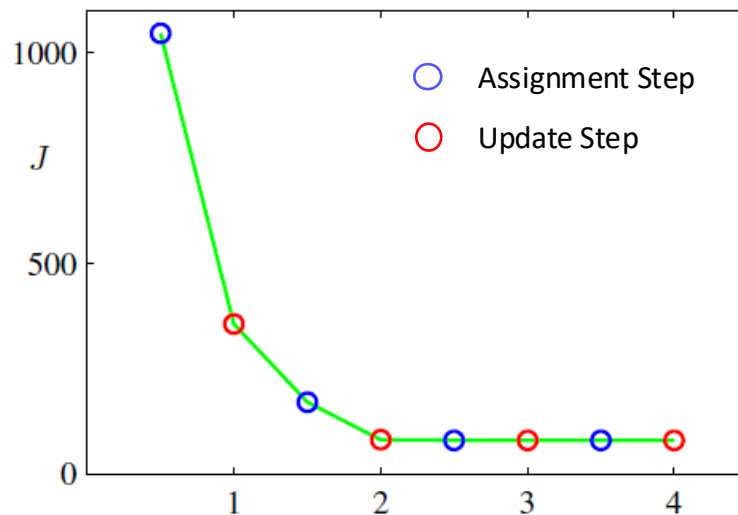
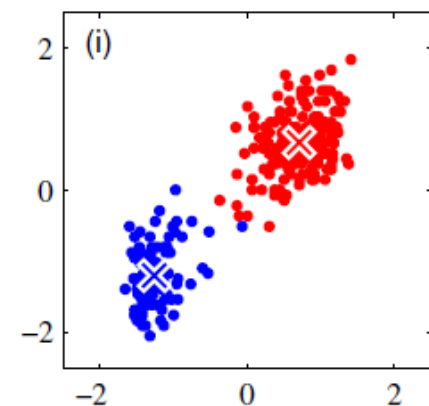
Iteration (4):

Assignment of r_{nk}



Iteration (4):

Update of μ_k



- Plot of the **cost function J** after each iteration.
- The algorithm has **converged** after the third update step.

Image source: "Pattern recognition and machine learning", Christopher Bishop

Application: Image Segmentation

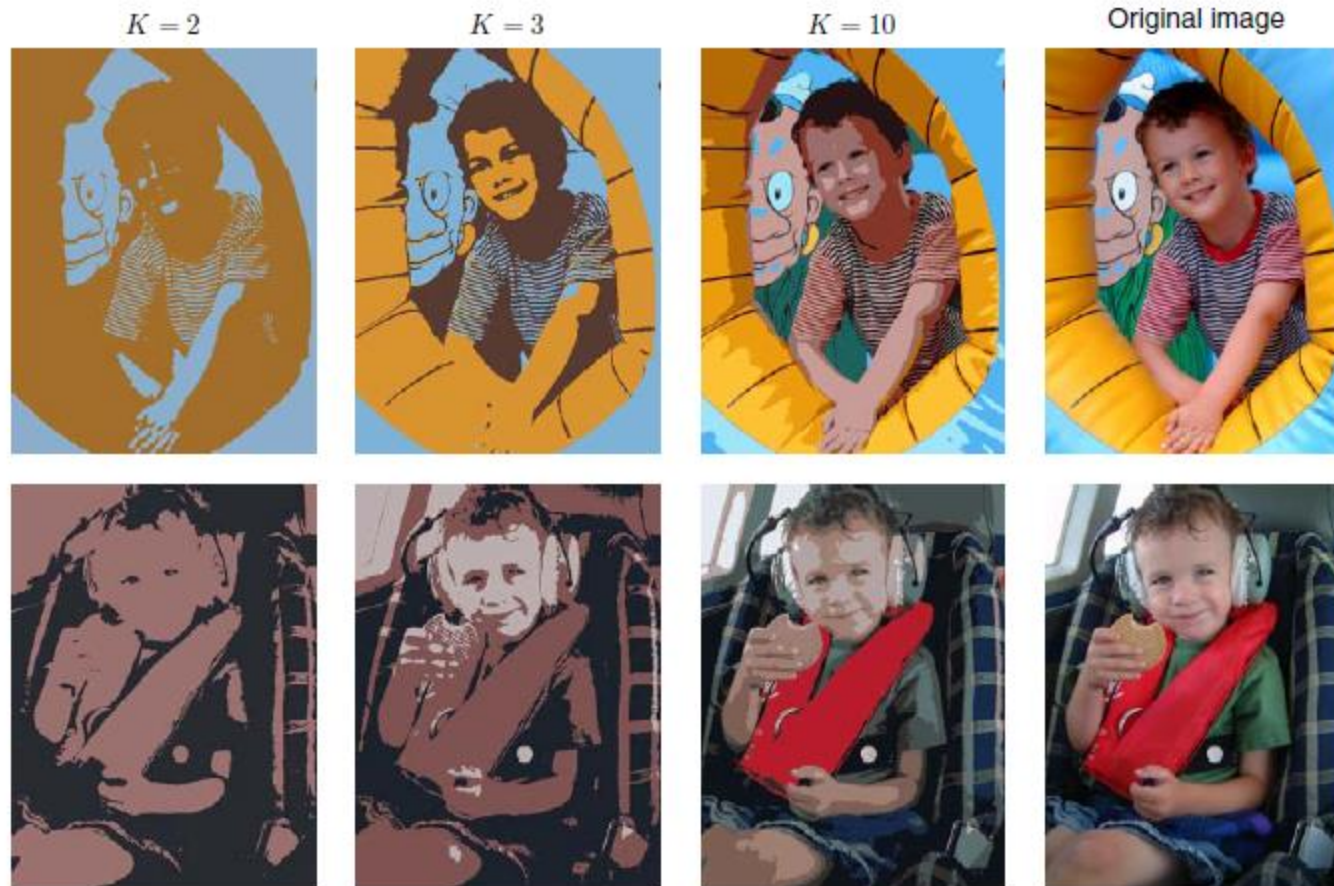


Image source: "Pattern recognition and machine learning", Christopher Bishop

Probabilistic Approach

Can we model the problem with a probabilistic approach?

Gaussian Mixture Models

- **“Old Faithful” dataset**: 272 measurements of the eruption of the Old Faithful geyser at Yellowstone National Park, USA.
- Data set forms two dominant clumps, a simple **Gaussian distribution is unable to capture** this structure.
- A **linear superposition of two Gaussians** gives a better characterization of the data set.



Photo source: “Old Faithful”, courtesy of Chen Li, June’18

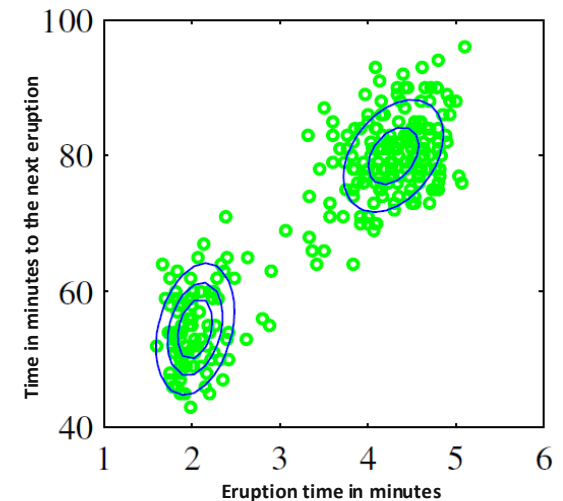
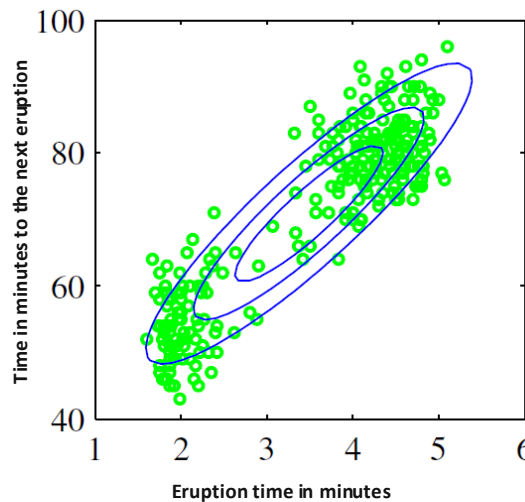
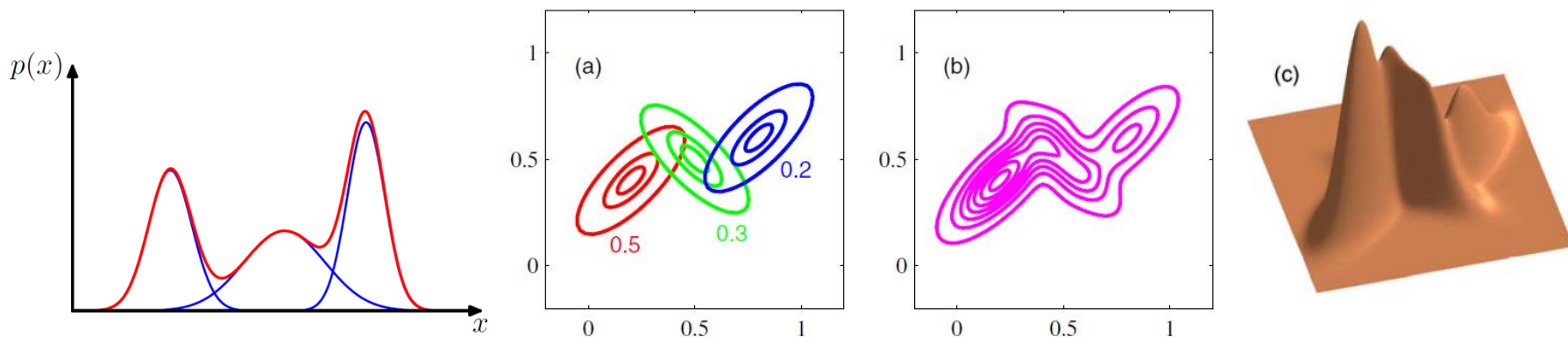


Image source: “Pattern recognition and machine learning”, Christopher Bishop

Gaussian Mixture Models

- **Mixture distributions**: Probabilistic models formed by taking linear combinations of more basic distributions such as Gaussian a.k.a **Gaussian Mixture Model (GMM)**.
- Linear combination of sufficient number of Gaussians give rise to very complex densities that can be used to **approximate almost any continuous density** with arbitrary accuracy.

Example:



1D GMM (red) formed by 3 Gaussians (blue)

(b)-(c) 2D GMM formed by (a) 3 Gaussians

Gaussian Mixture Models

- The **probability distribution** of a mixture of Gaussians is given by the superposition of K Gaussian densities:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) .$$

- Each Gaussian density $\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is called a **component of the mixture**, and has its own mean $\boldsymbol{\mu}_k$ and covariance $\boldsymbol{\Sigma}_k$.
- The parameters $0 \leq \pi_k \leq 1$ is the **mixing coefficients**, and must sum to one:

$$\sum_{k=1}^K \pi_k = 1 .$$

Gaussian Mixture Models

- Let us introduce a K -dimensional **binary random variable Z** having a **1-of- K representation**.
- $z_k = 1 \Rightarrow z_{j \neq k} = 0$ indicates the **assignment of the random variable x** to the k^{th} Gaussian density.
- The values of Z_k must satisfy:

$$z_k \in \{0,1\} \quad \text{and} \quad \sum_k z_k = 1$$

- **K possible states** for the vector Z according to which element is non-zero.

Gaussian Mixture Models

- The marginal distribution of Z is a **categorical distribution** specified in terms of the **mixing coefficients** π_k :

$$p(z) = \prod_{k=1}^K \pi_k^{z_k} = \text{cat}_z[\pi] \ ,$$

where the parameter $\pi = [\pi_1, \dots, \pi_K]$ must be:

$$0 \leq \pi_k \leq 1 \quad \text{and} \quad \sum_{k=1}^K \pi_k = 1 \quad .$$

Gaussian Mixture Models

- **Conditional distribution** of X given a particular value for Z is a Gaussian:

$$p(\mathbf{x}|z_k = 1) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

- Which can also be written as:

$$p(\mathbf{x}|\mathbf{z}) = \prod_{k=1}^K \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k}.$$

Gaussian Mixture Models

- The **joint distribution** $p(\mathbf{x}, \mathbf{z})$ is given by the following DGM:

$$p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z})p(\mathbf{x}|\mathbf{z})$$



- The **marginal distribution of X** is then obtained by summing the joint distribution over all possible states of the **latent variable Z** to give:

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z}) = \sum_{\mathbf{z}} \prod_{k=1}^K \pi_k^{z_k} \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)^{z_k}$$

Gaussian Mixture Models

$$\begin{aligned} p(\mathbf{x}) &= \sum_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x}|\mathbf{z}) \\ &= \sum_{\mathbf{z}} \prod_{k=1}^K \pi_k^{z_k} \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)^{z_k} \\ &= \underbrace{\left(\prod_{k=1}^K \pi_k^{z_k} \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)^{z_k} \right)_{z_{k=1}=1}}_{\pi_{k=1} \mathcal{N}(\mathbf{x} | \mu_{k=1}, \Sigma_{k=1})} + \dots + \\ &\quad \underbrace{\left(\prod_{k=1}^K \pi_k^{z_k} \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)^{z_k} \right)_{z_{k=K}=1}}_{\pi_{k=K} \mathcal{N}(\mathbf{x} | \mu_{k=K}, \Sigma_{k=K})} \\ &= \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k) \end{aligned}$$

Gaussian Mixture Models

- Another quantity that will play an important role is $p(z_k = 1 | \mathbf{x})$ denoted as $\gamma(z_k)$, whose value can be found using **Bayes' theorem**:

$$\begin{aligned}\gamma(z_k) \equiv p(z_k = 1 | \mathbf{x}) &= \frac{p(z_k = 1)p(\mathbf{x} | z_k = 1)}{\sum_{j=1}^K p(z_j = 1)p(\mathbf{x} | z_j = 1)} \\ &= \frac{\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}\end{aligned}$$

Prior probability
of $z_k = 1$



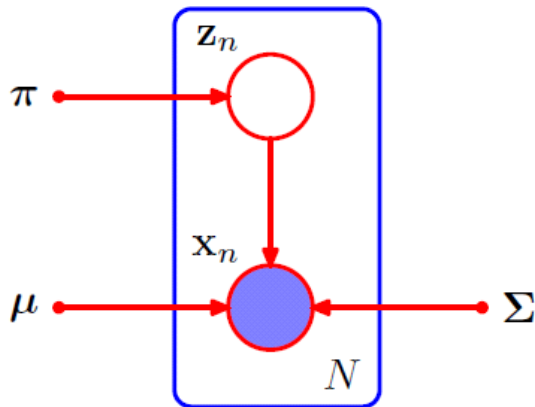
- As we shall see later, $\gamma(z_k)$ can also be viewed as the **responsibility** that component k takes for 'explaining' the observation $\mathbf{x}$.

Maximum Log-Likelihood

- Suppose we have a data set of **N i.i.d. observations** $\{x_1, \dots, x_N\}: x_n \in \mathbb{R}^D$, we model the **log-likelihood** as:

$$\ln p(x_1, \dots, x_N | \theta) = \sum_{n=1}^N \ln \underbrace{\sum_{z_n} p(x_n, z_n | \theta)}_{\text{Unknown parameter } \theta}$$

Incomplete data because $Z = \{z_1, \dots, z_N\} \in \mathbb{R}^{N \times K}$ is a latent variable



$$= \sum_{n=1}^N \ln \sum_{z_n} p(z_n | \pi) p(x_n | z_n, \mu, \Sigma)$$

$$= \sum_{n=1}^N \ln \sum_{k=1}^K \pi_k \mathcal{N}(x_n | \mu_k, \Sigma_k),$$

where

$$\theta = \{\pi_1 \dots \pi_K, \mu_1 \dots \mu_K, \Sigma_1 \dots \Sigma_K\}.$$

Maximum Log-Likelihood

$$\operatorname{argmax}_{\theta} \ln p(\mathbf{x}_1, \dots, \mathbf{x}_N | \theta) = \operatorname{argmax}_{\pi, \mu, \Sigma} \sum_{n=1}^N \ln \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)$$

- The **summation term inside the logarithm** prevents the logarithm function from acting directly on the Gaussian.
- We shall see that we will **no longer obtain a closed-form solution** of the unknown parameters by setting the derivatives to zero!

Maximum Log-Likelihood

- Setting the derivatives of $\ln p(\mathbf{x}_1, \dots, \mathbf{x}_N | \theta)$ w.r.t. μ_k of the Gaussian components to zero, we obtain:

$$0 = - \sum_{n=1}^N \underbrace{\frac{\pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n | \mu_j, \Sigma_j)}}_{\gamma(z_{nk})} \Sigma_k^{-1} (\mathbf{x}_n - \mu_k)$$

- Multiplying by Σ_k (which we assume to be non-singular) and rearranging, we obtain:

$$\mu_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n, \quad \text{where} \quad N_k = \sum_{n=1}^N \gamma(z_{nk})$$

Not closed-form since responsibility is a function of π_k, μ_k, Σ_k !

Maximum Log-Likelihood

- If we set the derivative of $\ln p(\mathbf{x}_1, \dots, \mathbf{x}_N | \theta)$ w.r.t. Σ_k to zero, we get:

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^T$$

Not closed-form since responsibility is a function of π_k, μ_k, Σ_k !

Maximum Log-Likelihood

- Finally, we maximize $\ln p(\mathbf{x}_1, \dots, \mathbf{x}_N | \theta)$ w.r.t. π_k subjected to $\sum_k \pi_k = 1$ by maximizing the following auxiliary equation:

$$\ln p(\mathbf{X} | \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

Lagrange multiplier

- Which gives:

$$0 = \sum_{n=1}^N \underbrace{\frac{\mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}}_{= N_k / \pi_k} + \lambda$$

- Multiply both sides by π_k and sum over k making use of the constraint $\sum_k \pi_k = 1$, we get:

$$\sum_{n=1}^N \frac{\sum_k \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} = -\lambda \sum_k \pi_k \quad \Rightarrow \quad \lambda = -N$$

Maximum Log-Likelihood

- Using $\lambda = -N$ to eliminate λ and rearranging, we get:

$$\pi_k = \frac{N_k}{N}, \quad \text{where} \quad N_k = \sum_{n=1}^N \gamma(z_{nk})$$

Not closed-form since responsibility is a function of π_k, μ_k, Σ_k !

- This is the **average responsibility** which the k^{th} component takes for explaining the data points.

Maximum Log-Likelihood

- The maximum log-likelihood estimates of the unknown parameter **do not constitute a closed-form solution** because of the responsibilities $\gamma(z_{nk})$.
- However, these results do suggest a **simple iterative scheme** for finding a solution to the maximum likelihood problem!

EM for Gaussian Mixtures

Given a Gaussian mixture model, the goal is to **maximize the likelihood function** w.r.t. the parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$.

1. **Initialize** the means μ_k , covariances Σ_k and mixing coefficients π_k , and **evaluate** the initial value of the log likelihood.
2. **Expectation Step**: Evaluate the **responsibilities** $\gamma(Z)$ using the current parameter values

$\gamma(Z)$ is a $N \times K$ table where each entry is $\gamma(z_{nk})$

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

EM for Gaussian Mixtures

Given a Gaussian mixture model, the goal is to **maximize the likelihood function** w.r.t. the parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$.

3. **Maximization Step**: Re-estimate the **parameters** using the current responsibilities

$$\begin{aligned}\mu_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n \\ \Sigma_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}}) (\mathbf{x}_n - \mu_k^{\text{new}})^T \\ \pi_k^{\text{new}} &= \frac{N_k}{N}\end{aligned}$$

where

$$N_k = \sum_{n=1}^N \gamma(z_{nk}).$$

EM for Gaussian Mixtures

Given a Gaussian mixture model, the goal is to **maximize the likelihood function** w.r.t. the parameters $\theta = \{\pi_k, \mu_k, \Sigma_k\}$.

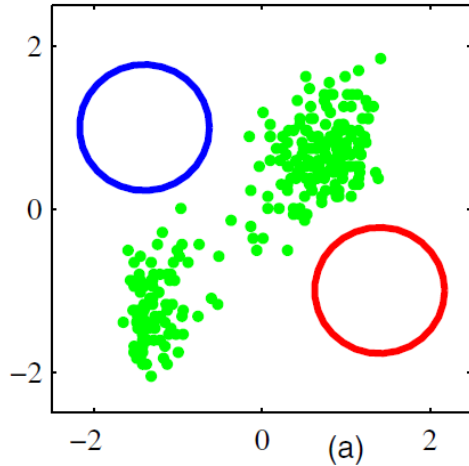
4. Evaluate the **log likelihood**:

$$\ln p(\mathbf{X}|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

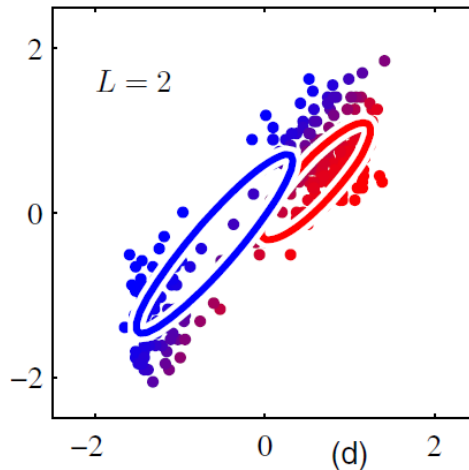
and **check for convergence** of either the parameters or the log likelihood. If the convergence criterion is NOT satisfied return to step 2.

Illustration of the EM Algorithm

Initialization: random μ_k ,
identity Σ_k and $\pi_k = 0.5$

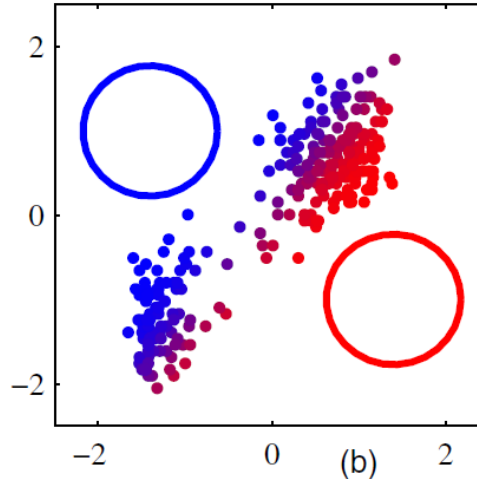


M Step: 2nd iteration

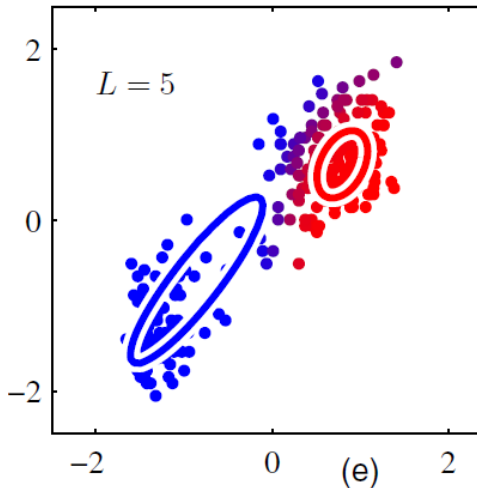


E Step: 1st iteration

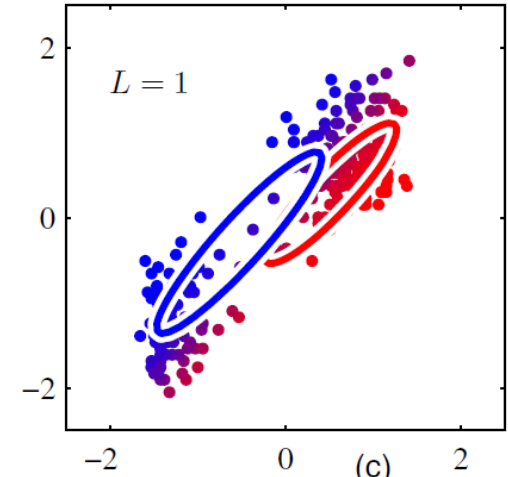
(color of dots used to illustrate the
strength of the assignment posterior)



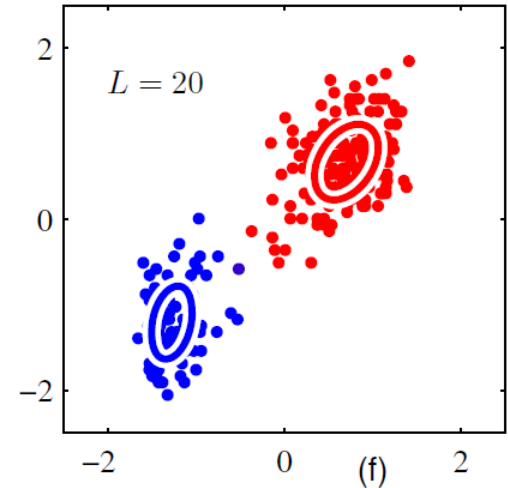
M Step: 5th iteration



M Step: 1st iteration



M Step: 20th iteration



EM for GMM: Initialization

- Run the K -means algorithm first to **find a suitable initialization** for a GMM that is subsequently adapted using EM.

Remarks:

- EM algorithm takes **many more iterations** to reach (approximate) convergence compared with the K -means algorithm.
- And each cycle requires **significantly more computation**.

The General EM Algorithm

- The goal of the EM algorithm is to find **maximum likelihood solutions** for models having **latent variables**.
- The log-likelihood function is given by:

$$\ln p(\mathbf{X}|\theta) = \ln \left\{ \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta) \right\} \quad \text{or} \quad \ln p(X|\theta) = \ln \left\{ \int_{\mathbf{Z}} p(X, \mathbf{Z}|\theta) \right\}$$

 Discrete latent variable Continuous latent variable 

where

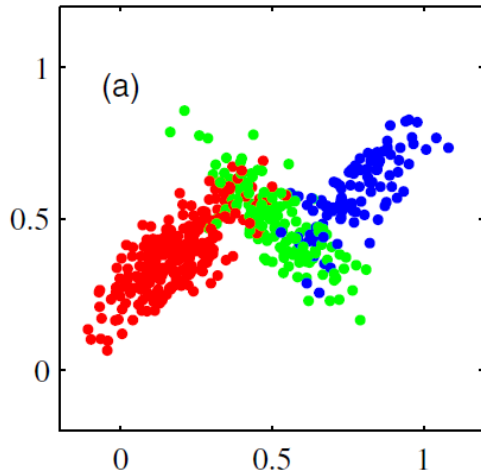
- $\mathbf{X}$: set of all **observed data** with n^{th} row represents $\mathbf{x}_n^T$
- $\mathbf{Z}$: set of all **latent variables** with corresponding row $\mathbf{z}_n^T$
- θ : set of all **model parameters**

The General EM Algorithm

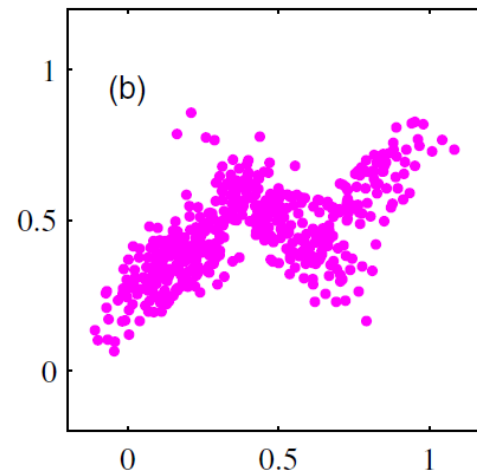
$$\ln p(\mathbf{X}|\boldsymbol{\theta}) = \ln \left\{ \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta}) \right\} \quad \text{or} \quad \ln p(X|\theta) = \ln \left\{ \int_{\mathbf{Z}} p(X, \mathbf{Z}|\theta) \right\}$$

- A key observation is that the summation/integration over the latent variables **appears inside the logarithm**.
- Marginal distribution $p(X|\theta)$ **does not simplify** even if the joint distribution $p(X, \mathbf{Z}|\theta)$ belongs to the exponential family, e.g. Gaussian.
- Resulting in **complicated (non closed-form) expressions** for the maximum likelihood solution.

The General EM Algorithm



Assignment of x_n into clusters is known: Complete data



Assignment of x_n into clusters is unknown: Incomplete data

- **Complete data:** Both observation X and latent variable Z are known.
- **Incomplete data:** Observation X is known, but latent variable Z is unknown.

The General EM Algorithm

- It is straightforward to maximize the likelihood function for the **complete data set** $\ln p(X, Z|\theta)$.
- In practice, however, we are **not given** the complete data set $\{X, Z\}$, but only the incomplete data X .
- Instead, we consider **maximization** of the **expected value** of the complete data log-likelihood $\ln p(X, Z|\theta)$ w.r.t. $p(Z|X, \theta)$.
- Do the Expectation and Maximization steps **iteratively** until convergence.

The General EM Algorithm

Expectation Step:

- Use the **current parameter values** θ^{old} to find the posterior distribution of the latent variables given by $p(\mathbf{Z}|\mathbf{X}, \theta^{old})$.
- We then use $p(\mathbf{Z}|\mathbf{X}, \theta^{old})$ to find the **expectation of the complete-data log likelihood** evaluated for some general parameter value θ .
- This expectation, denoted $Q(\theta, \theta^{old})$, is given by:

$$\begin{aligned} Q(\theta, \theta^{old}) &= \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta). \\ &= \mathbb{E}_{\mathbf{Z}|\mathbf{X}, \theta^{old}} [\ln p(\mathbf{X}, \mathbf{Z}|\theta)] \end{aligned}$$

The General EM Algorithm

Maximization Step:

- Determine the **revised parameter estimate** θ^{new} by maximizing this function:

$$\begin{aligned}\theta^{new} &= \arg \max_{\theta} Q(\theta, \theta^{old}) \\ &= \arg \max_{\theta} \underbrace{\sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta)}\end{aligned}$$

Log is now inside the summation!

- Since the logarithm now acts directly on the joint distribution $p(\mathbf{X}, \mathbf{Z}|\theta)$, the corresponding M-step will be in **closed-form**.

The General EM Algorithm

- **Given:** a **joint distribution** $p(X, Z|\theta)$ over observed variables X and latent variables Z , governed by parameters θ .
- **Goal:** is to **maximize the likelihood function** $p(X|\theta)$ with respect to θ .

The General EM Algorithm

1. Choose an **initial setting** for the parameters θ^{old} .

2. **Expectation step**: Evaluate $p(\mathbf{Z}|\mathbf{X}, \theta^{old})$.

3. **Maximization step**: Evaluate θ^{new} given by:

$$\theta^{new} = \arg \max_{\theta} Q(\theta, \theta^{old})$$

where

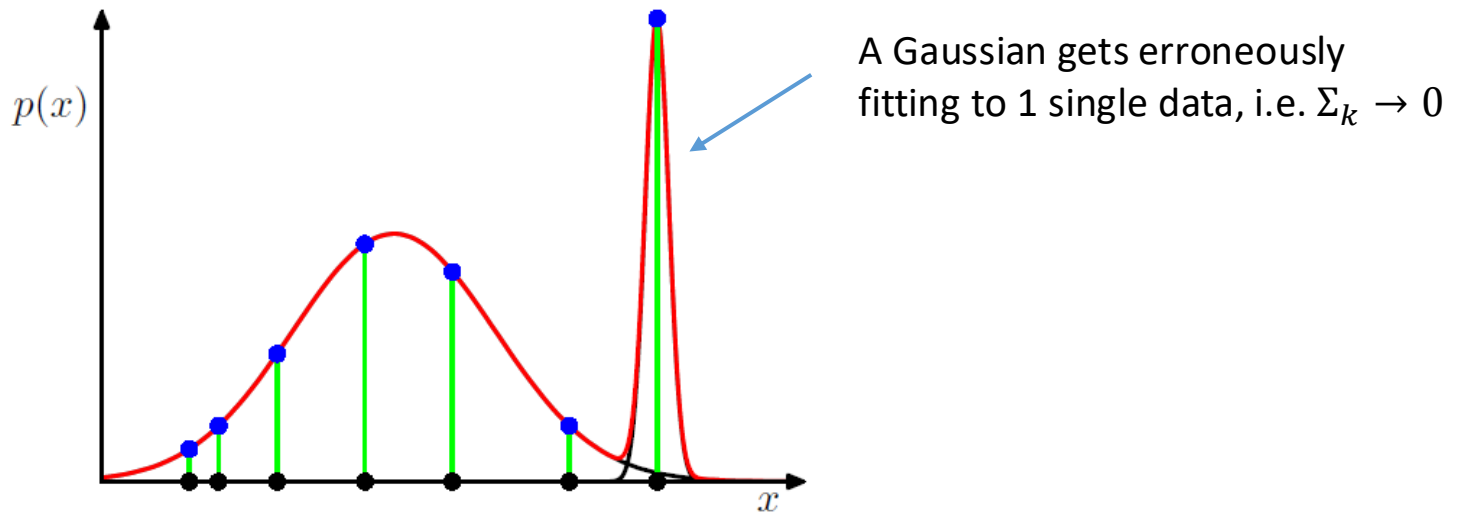
$$Q(\theta, \theta^{old}) = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta)$$

4. Check for convergence of either the log likelihood or the parameter values, **if not converged**:

$$\theta^{old} \leftarrow \theta^{new}$$

Singularities and MAP

Illustration of the “singularity problem”:



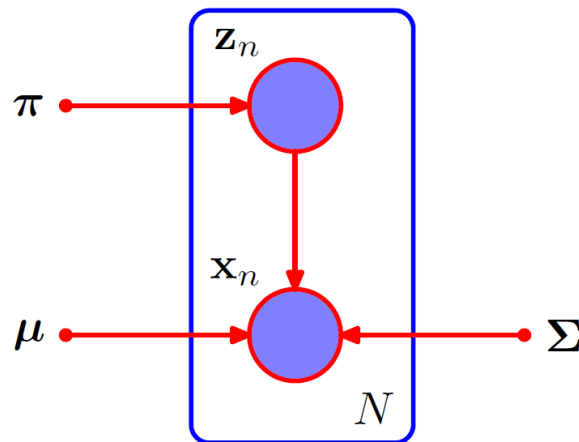
- Problem can be alleviated by applying MAP on $Q(\theta, \theta^{old})$.

$$Q(\theta, \theta^{old}) + \ln p(\theta) \leftarrow \text{Dirichlet prior on } \pi_k, \text{ and normalized inversed Gaussian on } (\mu_k, \sigma_k).$$

Gaussian Mixture Revisited

- The **complete data log-likelihood** is given by:

$$\ln p(\mathbf{X}, \mathbf{Z} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) = \ln \prod_{n=1}^N \prod_{k=1}^K \pi_k^{z_{nk}} \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_{nk}}$$



$$= \sum_{n=1}^N \sum_{k=1}^K z_{nk} \underbrace{\{\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\}}$$

Now the log is inside the summation!

Gaussian Mixture Revisited

Complete data log-likelihood:

$$\ln p(X, Z \mid \pi, \mu, \Sigma) = \sum_{k=1}^K \sum_{n=1}^N z_{nk} (\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n \mid \mu_k, \Sigma_k))$$

- We can **switch the sums** and only look at the points associated with the respective mixture component k .
- Maximization of each (μ_k, Σ_k) can be **done separately**, similarly we can solve for π_k by enforcing the sum to one constraint using Lagrange multiplier.
- **Problem:** the latent variables Z are **not observed!!!**

Gaussian Mixture Revisited

- Instead of maximizing the joint log-likelihood, we maximize its **expectation** under the latent variable distribution:

$$\begin{aligned}\mathbb{E}_Z[\ln p(X, Z|\theta)] &= \sum_Z p(Z|X, \theta^{old}) \ln p(X, Z|\theta) \\&= \sum_n \sum_k p(z_{nk}|x_n, \theta^{old}) \ln p(x_n, z_{nk}|\theta) \\&= \sum_n \sum_k \frac{p(x_n|z_{nk}, \theta^{old})p(z_{nk}|\theta^{old})}{p(x_n|\theta^{old})} \ln p(x_n, z_{nk}|\theta) && \text{(Bayes Rule)} \\&= \sum_n \sum_k \underbrace{\frac{\mathcal{N}(x_n|\mu_k^{old}, \Sigma_k^{old})\pi_k^{old}}{\sum_j \mathcal{N}(x_n|\mu_j^{old}, \Sigma_j^{old})\pi_j^{old}}}_{\gamma(z_{nk})} \ln p(x_n, z_{nk}|\theta) \\&\quad \gamma(z_{nk}) \quad \text{Responsibility that stays fixed} \\&= \sum_n \sum_k \gamma(z_{nk}) [\ln \pi_k + \ln \mathcal{N}(x_n|\mu_k, \Sigma_k)]\end{aligned}$$

Gaussian Mixture Revisited

$$\mathbb{E}_Z[\ln p(X, Z|\theta)] = \sum_k \sum_n \gamma(z_{nk}) [\ln \pi_k + \ln \mathcal{N}(x_n|\mu_k, \Sigma_k)]$$

- Similar to the complete log-likelihood case, we can **switch the sums** and only look at the points associated with the respective mixture component k .
- Since $\gamma(z_{nk})$ stays fixed, maximization of each (μ_k, Σ_k) can be **done separately**, similarly we can solve for π_k by enforcing the sum to one constraint using Lagrange multiplier.

The Theory Behind EM Algorithm

- Maximizing the log-likelihood $\ln p(X|\theta)$ is difficult due to the **need of marginalizing over the latent variables** inside the logarithm:

$$\ln p(X|\theta) = \ln \sum_Z p(X, Z|\theta)$$

- It turns out that $\ln p(X|\theta)$ can be maximized by **maximizing its lower bound** $\mathcal{L}(q, \theta)$ within the EM algorithm.

The Theory Behind EM Algorithm

- Let us rewrite the log-likelihood into:

$$\ln p(\mathbf{X}|\boldsymbol{\theta}) = \underbrace{\mathcal{L}(q, \boldsymbol{\theta})}_{\text{Lower bound}} + \underbrace{\text{KL}(q||p)}_{\text{KL Divergence}}$$

where

$$\mathcal{L}(q, \boldsymbol{\theta}) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\boldsymbol{\theta})}{q(\mathbf{Z})} \right\}$$

$$\text{KL}(q||p) = - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \underbrace{\frac{p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta})}{q(\mathbf{Z})}} \right\} \geq 0$$

$q(\mathbf{Z})$ is a distribution we defined over the latent variable

The Theory Behind EM Algorithm

Proof:

$$\begin{aligned}\mathcal{L}(q, \theta) + \text{KL}(q||p) &= \sum_Z q(Z) \ln \left\{ \frac{p(X, Z|\theta)}{q(Z)} \right\} - \sum_Z q(Z) \ln \left\{ \frac{p(Z|X, \theta)}{q(Z)} \right\} \\ &= \sum_Z q(Z) \{ \underbrace{\ln p(X, Z|\theta)}_{\ln p(\mathbf{Z}|\mathbf{X}, \boldsymbol{\theta}) + \ln p(\mathbf{X}|\boldsymbol{\theta})} - \cancel{\ln q(Z)} - \ln p(Z|X, \theta) + \cancel{\ln q(Z)} \} \\ &= \sum_Z q(Z) \ln p(X|\theta) \\ &= \ln p(X|\theta)\end{aligned}$$

□

The Theory Behind EM Algorithm

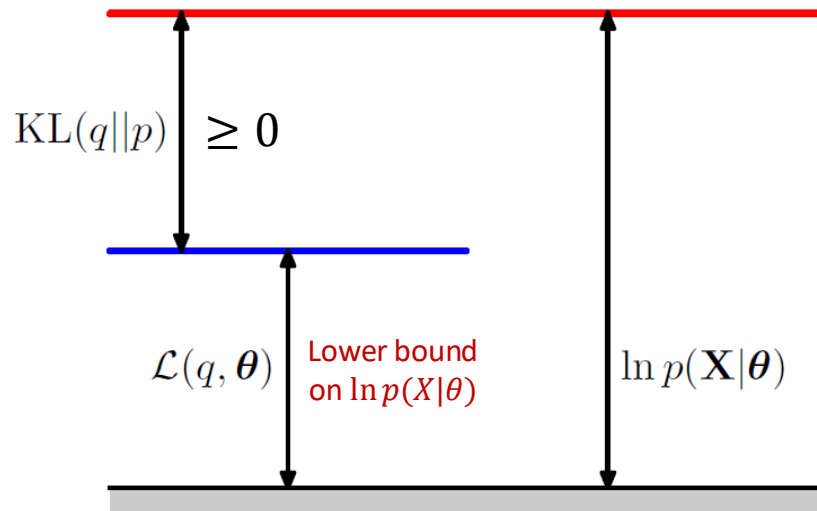


Illustration of the decomposition given by:

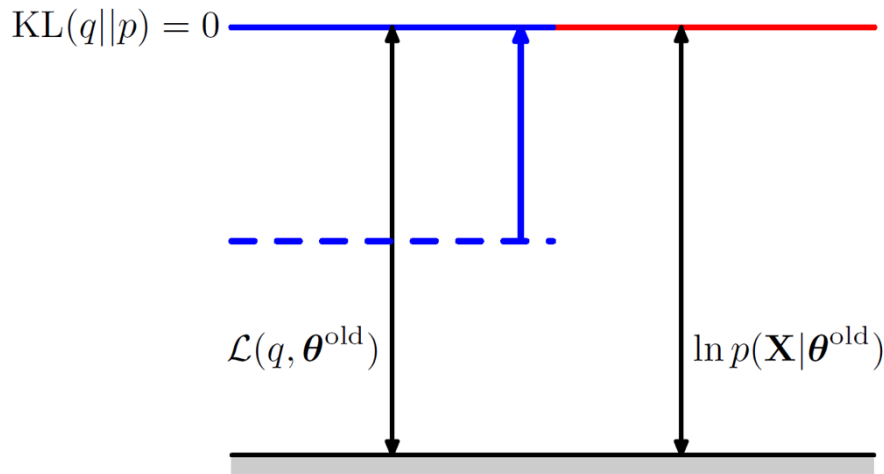
$$\ln p(\mathbf{X}|\theta) = \mathcal{L}(q, \theta) + KL(q||p)$$

which holds for any choice of $q(\mathbf{Z})$.

- Because the **Kullback-Leibler divergence** satisfies $KL(q \parallel p) \geq 0$, we see that the quantity $\mathcal{L}(q, \theta)$ is a **lower bound** on the log-likelihood function $\ln p(X|\theta)$.

The Theory Behind EM Algorithm

Illustration of the E-Step:



Lower bound $\mathcal{L}(q, \theta)$ is maximized by choosing $q(Z) = p(Z|X, \theta^{old})$!

Proof:

$$\begin{aligned} \text{KL}(q||p) &= - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{p(\mathbf{Z}|\mathbf{X}, \theta)} \right\} \\ &= 0 \end{aligned}$$

$\Rightarrow \mathcal{L}(q, \theta)$ must be at its maximum since

$$\ln p(\mathbf{X}|\theta) = \mathcal{L}(q, \theta) + \text{KL}(q||p)$$

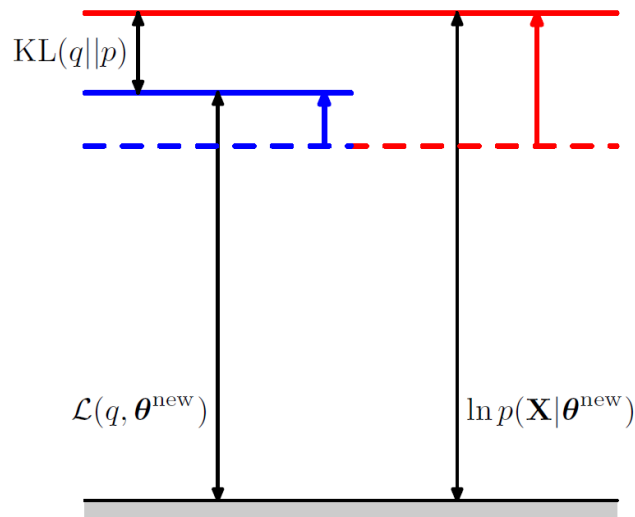
□

- Equivalence of **expectation** under the latent variable distribution:

$$\begin{aligned} \mathcal{L}(q, \theta) &= \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{X}, \mathbf{Z}|\theta) - \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \ln p(\mathbf{Z}|\mathbf{X}, \theta^{old}) \\ &= Q(\theta, \theta^{old}) + \text{const} \end{aligned}$$

The Theory Behind EM Algorithm

Illustration of the M-Step:

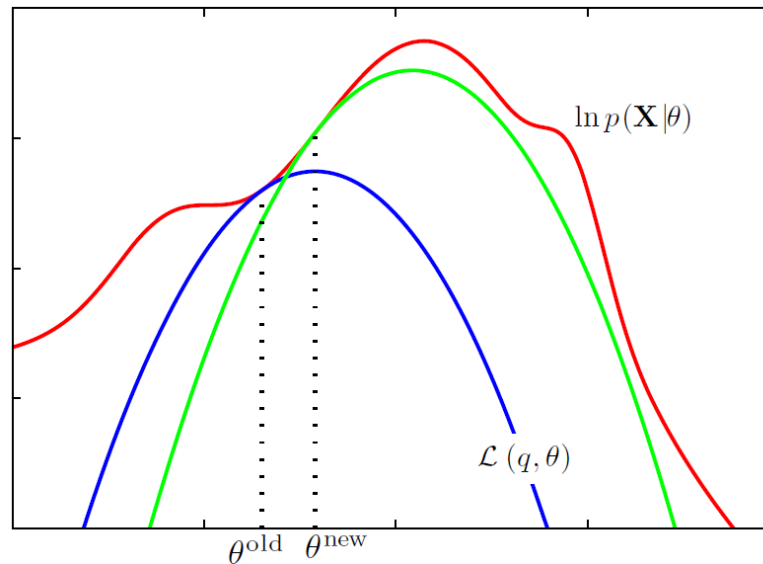


$$\mathcal{L}(q, \theta^{new}) = \sum_Z p(Z|X, \theta^{old}) \ln p(X, Z|\theta^{new}) + \text{const}$$

$$\text{KL}(q||p) = - \sum_Z p(Z|X, \theta^{old}) \ln \left\{ \frac{p(Z|X, \theta^{new})}{p(Z|X, \theta^{old})} \right\}$$

- The distribution $q(Z)$ is held fixed and the **lower bound $\mathcal{L}(q, \theta)$ is maximized** w.r.t θ to give a revised value θ^{new} .
- Because the KL divergence is nonnegative, this causes the log-likelihood $\ln p(X|\theta)$ **to increase** by at least as much as the lower bound does.

The Theory Behind EM Algorithm



- **E-step:** we compute the **convex lower bound** given the old parameters θ^{old} (blue curve).
- **M-step:** we **maximize this lower bound** to get new parameters θ^{new} .
- This is **repeated** (green curve) until convergence.

Summary

- We have looked at how to:
 1. Use the non-probabilistic **k-mean algorithm** to solve the clustering problem.
 2. Describe the **Gaussian-mixture model**.
 3. Apply the **Expectation-Maximization algorithm** for estimation of both the unknown parameters and latent variables.